MILITARY SPECIFICATION

CARTRIDGE, 25 MILLIMETER,

PGU-20/U (API)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the Cartridge, 25 Millimeter (mm), PGU-20/U Armor Piercing Incendiary (API), herein referred to as the Cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-12560	Armor Plate, Wrought, Homogeneous (For Use in and For Combat Vehicles Ammunition Testing).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Departments (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 1305

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-644	Visual Inspection Standard and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1167	Ammunition Data Card.
MIL-STD-1168	Lot Numbering of Ammunition.
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

OTHER GOVERNMENT DOCUMENTS

U.S. Army Material Command

AMCR 715-505	Ammunition Ballistic Acceptance List Methods, Volume 8, Test Procedures for 20 mm Cartridges.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1395AS125	Gun, Automatic, 25 mm (GAU-12/U).
1395AS160	Cartridge-Gun Interface, 25 mm.
1397AS190	Cartridge, 25 mm, API, PGU-20/U.
1397AS390	Cartridge, 25 mm, TP, PGU-23/U.

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PUBLICATIONS

Code of Federal Regulations

49 CFR 100-199	Hazardous Materials Regulations, Department of Transportation.

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402)

REPORTS

BAL TR No. 66	Resistance of Steel Targets to Perforation by Small Caliber Armor Piercing Projectiles.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The cartridge shall comply with Drawing 1397AS190 and other requirements specified herein.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Chamber pressure. The average plus three standard deviations of the peak chamber pressure shall be not greater than 425 magapascals (MPa) at 21 ±3 degrees Celsius (°C). The average plus three standard deviations of peak chamber pressure shall be not greater than 476 MPa when functioning at any individual temperature between -54 and 71°C.

3.2.1.2 Action time. The time between firing-pin contact with the primer and exit of the projectile from the muzzle shall be not greater than 6.0 milliseconds (ms) at any cartridge temperature from -54 to 71°C.

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3.2.1.3 Velocity. The average projectile muzzle velocity of cartridges conditioned at 21 ±3°C shall be a minimum of 1,000 meters per second (m/s). The standard deviation shall be a maximum of 10 m/s. At any cartridge temperature from -54 to 71°C, the projectile muzzle velocity shall be greater than 950 m/s.

3.2.1.4 Accuracy. The impact dispersion of the projectiles shall be such that the standard deviation of the impact points about the mean impact point shall be not greater than 0.7 milliradians when fired in the single-shot test barrel at 21 ±3°C.

3.2.1.5 Projectile extraction. The force required to extract the projectile from the case shall be not less than of 8,000 newtons (N) and not greater than 16,900 N.

3.2.1.6 Projectile torque. The projectile shall not rotate in the case when subjected to a torque of 3.4 newton meters (Nm).

3.2.1.7 Penetration. The projectile shall exhibit a Probable Ballistic Limit (PBL) Velocity of not greater than 950 meters per second (m/s) when fired against 38.1 mm of armor (conforming to MIL-S-12560) at an obliquity of 45 degrees to the flight path of the projectile.

3.2.1.8 Structural integrity. The cartridge shall be structurally compatible with the GAU-12/U gun, Drawing 1395AS125. When fired, all parts of the projectile shall remain intact while in the gun and during flight.

3.2.1.9 Waterproofness. The mean velocity (wet) of the cartridge shall not vary from the mean velocity (dry) by more than 30 m/s when tested at 21°C. The action time shall meet the requirements of 3.2.1.2.

3.2.1.10 Function and casuality. The cartridge shall function in the AV-8/25 mm Gun System, without any firing defects as specified in Table I.

Table I. Firing defect. 1/

Item	Defect	AQL
Cartridge	Premature Fire 2/	0.065
	Hangfire 3/	0.065
	Misfire 4/	0.065
	Failure to chamber 5/	0.065
	Failure to extract 6/	0.065
	Projectile remains in bore 7/	0.065

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Table I. Firing defect. 1/ - (Continued)

Item	Defect	AQL
Primer	Pierced primer 8/	0.65
	Blown primer 9/	0.40
	Primer leak 10/	0.40
	Loose primer 11/	0.65
Cartridge Case	Split or rupture (mouth, neck, shoulder) 12/	0.65
	Split or rupture (main body) 13/	0.40
	Split or rupture (base area) 14/	0.065
	Complete rupture 15/	0.065
	Detached metal 16/	0.250
Projectile	Parts separation and integrity 17/	0.065
	Rotating band separation 18/	0.25
	Projectile Yaw 19/	1.0

Table I Notes:

1/ These defects apply only to cartridges fired in the approved test fixture. Unless otherwise specified herein, firing defects shall be as defined in AMCR 715-505, Volume 8, Chapter 1, Section 3. The AQLs are from Table V and Table III.

2/ Premature fire. The cartridge functions prior to insertion in the chamber of the test fixture or gun.

3/ Hangfire. An action time in excess of 6 ms.

4/ Misfire. If a misfire occurs, a second firing-pin impact is not allowed unless a light firing-pin impact is confirmed by inspection. The cartridge configuration and firing conditions shall be recorded, and the cartridge and record submitted to failure analysis as defined by the contractor. Disposition of the lot shall be dependent upon results of the failure analysis. The failure analysis shall determine which of the following two misfire conditions apply:

a. Misfire due to primer failure. A primer misfire is defined as a normally chambered cartridge in which the primer fails to fire upon application of the prescribed energy to the primer. The failure shall be recorded and the procuring activity shall be notified and the lot accepted by the procuring activity (see 6.2.2).

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Table I Notes: (Continued)

b. Misfire not due to primer. A non-primer caused misfire is defined as a normally chambered cartridge in which the propellant does not ignite when the primer fires normally. Disposition of the lot will be as specified in the contract or purchase order (see 6.2.1).

5/ Failure to chamber. The cartridge cannot be inserted into the chamber a sufficient distance to allow the breech to be engaged.

6/ Failure to extract. The fired case can not be removed from the test weapon chamber by normal means.

7/ Projectile remains in bore. The projectile remains in the bore of the test weapon after firing.

8/ Pierced primer. The cartridge case shall be visually examined. A pierced primer is one in which the indentation in the primer cup, made by the firing pin, is entirely perforated. It is identified by a visual hole through the primer, or if the perforation is minute, by discoloration of the indentation caused by burning gases.

9/ Blown primer. The cartridge case shall be visually examined. A blown primer is a primer that when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and the primer pocket are enlarged and deformed.

10/ Primer leak. Primer leaks are defined as follows:

a. Primer leak, minor. Discoloration caused by gas leakage around the junction between the primer cup and primer pocket wall encompassing 180 degrees or more of the periphery. The lot shall not be penalized if this condition is observed; however, this condition shall be recorded. An occurrence rate of more than 10 percent of the sample shall be cause for investigation and corrective action. It shall not be cause for lot rejection.

b. Primer leak, major. Visual evidence of metal erosion around the primer cup and primer pocket periphery that results in automatic gun system damage of such severity that it would cause a subsequent system malfunction.

11/ Loose primer. A looseness that permits the fired primer to fail from the primer pocket after the cartridge has been fired without distortion of the primer pocket. The cartridge case shall be visually examined.

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Table I Notes: (Continued)

12/ Cartridge case split or rupture (mouth, neck, shoulder). A split or rupture that exceeds five millimeter (mm) in any direction.

13/ Cartridge case split or rupture (main body). A split or rupture that exceeds five mm in any direction.

14/ Cartridge case split or rupture (base area). A split or rupture of any size in the base area not including primer leaks. The base area is the area of the case not supported by the chamber.

15/ Complete rupture. A separation of the cartridge case into two or more portions.

16/ Detached metal. Metal sheared or missing from the fired cartridge case exterior such as rim or neck shears shall be classed as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck in the crimped area.

17/ Parts separation and integrity. A separation of the windscreen from the projectile or a projectile breakup shall be classed as a defect.

18/ Rotating band separation. A separation of a complete rotating band from the projectile shall be classed as a defect. Rotating band fragments having a length less than 10 mm are acceptable providing the band seat serration imprints are not visible.

19/ Projectile yaw. Defined as yaw angle greater than 15 degrees as determined by yaw/witness cards or a missed target during lot acceptance test. All functional tests specified in Table I and Table II (when required) shall be monitored.

3.2.2 Environmental conditions. The cartridge shall meet the performance requirements of this specification after having been subjected to the environments specified in 3.2.2.1 through 3.2.2.3.

3.2.2.1 Temperature shock. The cartridge shall meet the performance requirements of this specification after exposure to the temperature shock specified in MIL-STD-810, Method 503, Procedure I.

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Table II. First article inspection.

First Article Sampling				
No.	Test	Sample Size	Requirement	Test
I	Projectile extraction	50	3.2.1.5	4.3.1.5
	Propellant weight 1/	50	3.3.2.2	4.3.1.10
II	Projectile torque	20	3.2.1.6	4.3.1.6
III	Waterproofness 2/	50	3.2.1.9	4.3.1.9
IV	Pressure -54, 71°C 3/, 4/	50	3.2.1.1	4.3.1.1
	Pressure 21°C 3/	50	3.2.1.1	4.3.1.1
	Velocity -54, 71°C 3/, 4/	50	3.2.1.3	4.3.1.3
	Velocity 21°C 3/	50	3.2.1.3	4.3.1.3
	Action Time -54, 71°C 3/, 4/	50	3.2.1.2	4.3.1.2
	Action Time 21°C 3/	50	3.2.1.2	4.3.1.2
V	Single-shot accuracy 5/	50	3.2.1.4	4.3.1.4
VI	Salt fog 6/	50	3.2.2.3	4.3.1.12.3
VII	Temperature shock 6/	50	3.2.2.1	4.3.1.12.1
VIII	Temperature, humidity 6/ and altitude	50	3.2.2.2	4.3.1.12.2
IX	Penetration 7/	30	3.2.1.7	4.3.1.7
X	Structural integrity 8/	Tests III thru X	3.2.1.8	4.3.1.8
XI	Function and casualty 8/	1,500	3.2.1.10	4.3.1.11.1

Table II Notes:

1/ The propellant shall be visually examined for contamination as specified in AMCR 715-505, Volume 8, and the propellant weight determined.

2/ The waterproofness sample shall be tested in the single-shot barrel at 21°C. The cartridge weight shall be recorded before and after immersion in water.

3/ The samples for pressure, velocity and action time at -54, 21, and 71°C may be combined.

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Table II Notes: (Continued)

4/ A sample size of 25 cartridges shall be used at both -54 and 71°C.

5/ The test sample for single-shot accuracy may be combined with the pressure, velocity and action time samples at 21°C.

6/ The salt fog, temperature-humidity-altitude and temperature shock samples shall be tested in the single-shot barrel at 21°C.

7/ Two PBLs shall be measured. Penetration test rounds are not included in structural integrity testing.

8/ Firing defects and associated criteria are specified in Table I. The accept/reject criteria shall be as specified in Table III.

Table III. Preproduction test function and casualty firing defect.

Accept/Reject Criteria (Single Sampling Plan)											
Test Number 1/	Description	Sample Size	Acceptable Quality Levels								
			0.065		0.25		0.40	0.65		1.0	2.5
			Acc/Rej		Acc/Rej		Acc/Rej	Acc/Rej		Acc/Rej	Acc/Rej
1 - 8	Single shot	350	0/1		2/3		4/5	6/7		8/9	16/17
9	Burst firing	1500	0/1		5/6		8/9	13/14		20/21	50/51

1/ The test numbers are the tests specified in Table II.

3.2.2.2 Temperature, humidity, and altitude. The cartridge shall meet the performance requirements of this specification after exposure to the temperature/humidity/altitude cycling conditions specified in MIL-STD-810, Method 518.

3.2.2.3 Salt fog. The cartridge shall meet the performance requirements of this specification after exposure to the salt sea atmosphere specified in MIL-STD-810, Method 509, Procedure I.

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3.2.3 Reliability. The cartridge shall have a performance reliability of not more than one failure per 250,000 rounds. The cartridge shall be considered to have failed for the purposes of measuring cartridge reliability, if any of the following occurs:

a. Premature ignition, hangfire or misfire (see Table I, Notes 2, 3 and 4).

b. Projectile structural failure.

c. Any failure of the cartridge that causes gun system failure.

3.3 Design and construction.

3.3.1 Production drawings. The cartridge shall be fabricated and assembled in accordance with Drawing 1397AS190.

3.3.1.1 Configuration. The cartridge, when assembled in accordance with Drawing 1397AS190 shall be compatible with the interface dimensions specified on Drawing 1395AS160.

3.3.2 Standards of manufacture.

3.3.2.1 Component inspection. The components used in the assembly of cartridges shall have been inspected and accepted in accordance with the applicable component drawings and the product fabrication specification referenced thereon.

3.3.2.2 Propellant weight. The propellant charge used in each cartridge shall be maintained within the weight limits necessary to meet the ballistic requirements of 3.2.1.1, 3.2.1.2 and 3.2.1.3.

3.3.2.3 Propellant contamination. The propellant shall contain not more than 0.10 percent foreign matter.

3.3.3 Lot identification. Unless otherwise specified herein, each cartridge and each packed ammunition lot shall be identified in accordance with the applicable drawings and MIL-STD-1168.

3.4 First article. When specified, a sample shall be subjected to first article inspection (see 4.1.2.1 and 6.2.3).

3.5 First article sample. The first article sample items shall be tested in accordance with Table II prior to regular production to verify the adequacy and suitability of processes and procedures used in the production of the item. A production sample shall be subjected to these tests whenever a change occurs in the manufacturing process, material used, drawings, specifications, or source of supply as to significantly affect product performance or uniformity. The first article lot size shall be a minimum of 2,500 cartridges.

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3.6 Workmanship. Parts shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for blemishes and light scratches. Required markings shall be neat and legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General. The tests defined herein are intended to verify that the manufacture and assembly procedures used in the production of the item do not degrade the item.

4.1.1.2 Test plan. The contractor shall provide the Government with a Production Quality Plan and a Lot Acceptance Test Plan 45 days prior to commencing to load, assemble, and pack (LAP) the first article or production lot of ammunition (see Table IV).

4.1.1.3 Test provisions.

4.1.1.3.1 Test conditions. Unless otherwise specified herein, all tests shall be conducted under local ambient conditions. If examination of data indicates an instrumentation/equipment failure, the test shall be declared "no test" and the test repeated.

4.1.1.3.2 Test measurement equipment. Tests specified herein shall be conducted using test equipment having a precision capability equal to or better than:

a.	Projectile velocity	±3 m/s
b.	Chamber pressure	±5.0 percent
c.	Action time	0.25 ms

d. Assembly and inspection gages shall comply with the cartridge/gun interface Drawing 1395AS160.

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e. All instrumentation used for ballistic data shall be calibrated in accordance with 4.1.1.3.4 and the data summary reported for both corrected and uncorrected results.

Table IV. Lot acceptance (cartridge tests).

Lot Acceptance Sampling					
No.	Test	Sample Size	Retest	Requirement	Test
I	Projectile extraction	35		3.2.1.5	4.3.1.5
	Propellant weight 1/	35		3.3.2.2	4.3.1.10
II	Projectile torque	10		3.2.1.6	4.3.1.6
III	Waterproofness 2/	20		3.2.1.8	4.3.1.9
IV	Pressure -54, 71°C 3/, 4/	25	25	3.2.1.1	4.3.1.1
	Pressure 21°C 3/	100	150	3.2.1.1	4.3.1.1
	Velocity -54, 71°C 3/, 4/	25	25	3.2.1.3	4.3.1.3
	Velocity 21°C 3/	100	150	3.2.1.3	4.3.1.3
	Action Time -54, 71°C 3/, 4/	25	25	3.2.1.2	4.3.1.2
	Action Time 21°C 3/	100	150	3.2.1.2	4.3.1.2
V	Single-shot accuracy 5/	20		3.2.1.4	4.3.1.4
VI	Structural integrity 6/	Test IV		3.2.1.7	4.3.1.8
VII	Function and casualty 6/	150		3.2.1.9	4.3.1.11.2

Table IV Notes:

1/ The propellant shall be visually examined for contamination as specified in AMCR 715-505, Volume 8 and the propellant weight determined.

2/ The waterproofness sample shall be tested in the single-shot barrel at 21°C. The cartridge weight shall be recorded before and after immersion in water.

3/ The samples for pressure, velocity and action time at -54, 21, and 71°C may be combined.

4/ A sample size of 25 cartridges shall be used at both -54 and 71°C.

5/ The test sample for single-shot accuracy may be combined with the pressure, velocity and action time samples at 21°C. Twenty consecutive cartridges shall be scored.

6/ Firing defects and associated criteria are specified in Table I. The accept/reject criteria shall be as specified in Table V.

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4.1.1.3.4 Calibration of instrumentation. A lot of PGU-23/U 25 mm target practice cartridges will be furnished by the procuring activity for use in establishing a reference baseline for testing the API cartridge specified herein. Twenty PGU-23/U cartridges from the reference lot will be fired at 21°C as a reference baseline. The cartridge will be fired from the standard barrel and test stand and the accuracy data recorded as specified in 4.3.1.4. The muzzle velocity, pressure and action time will also be recorded. The data will be compared with the original lot acceptance test (LAT) data of the reference lot. A baseline (X_B) pressure, velocity, and action time will be established. This baseline will require consultation and approval of the Government.

4.1.1.3.4.1 Daily reference. During the LAT of the API cartridge a minimum of 10 cartridges per day shall be fired to establish a daily reference (X_D) pressure (case mouth), velocity and action time.

4.1.1.3.4.2 Ballistic variation. Ballistic variation (ΔX) for pressure, velocity and action time between the reference baseline and the daily reference shall be calculated as follows:

$$\Delta X = X_B - X_D$$

The computed ballistic variation (ΔX) for pressure, velocity, and action time shall be utilized in 4.3.1.1, 4.3.1.2, and 4.3.1.3 to establish corrected performance.

4.1.1.3.5 Test fixture. Functional tests shall be conducted in a single-shot test fixture, using a single-shot test barrel.

4.1.1.3.6 Defective weapons. If any sample cartridge firing defect is found to have been caused by a defective test fixture, the firing defect shall not be classed as a failure. The defective test fixture shall be corrected or replaced and the test repeated in whole or in part, as appropriate. If the firing defect is not found to have been caused by a defective fixture, it shall be classed as a failure.

4.1.1.3.7 Test data. Test data shall be recorded in uncorrected form as well as after correction for reference round data in accordance with the calibration procedure of 4.1.1.3.4. The test data shall also include any anomalies that may occur during all preproduction and lot acceptance tests. The standard deviation data shall be recorded on the format shown in Figure 1, or equivalent.

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4.1.1.3.8 Reference rounds. Reference rounds shall be fired in conjunction with ballistic and accuracy tests. Unless otherwise specified herein, the reference rounds shall be assembled in accordance with the requirements of Drawing 1397AS390.

4.1.1.3.8.1 Ballistic reference rounds. A minimum of 10 reference rounds shall be fired during each ballistic firing test and the mean muzzle velocity, case mouth pressure and action time recorded. Any individual round readings which vary from the mean by greater than 15 m/s, 30 MPa, or 0.35 ms respectively, shall be disregarded and a new mean value shall be computed. If more than three rounds are disregarded, additional reference rounds shall be fired, provided they are fired during the same day as the applicable tests are conducted. The differences between the mean reference round test values and the reference acceptance baseline values shall not be greater than 10 m/s, 25 MPa, or 0.35 ms. Reference round data shall be utilized as defined in 6.4.

4.1.1.3.9 Unlisted firing defects. If a malfunction or casualty not covered by this specification occurs in any firing test, the lot shall be suspended and held for disposition as specified in the contract or purchase order (see 6.2.1).

4.1.2 Special tests and examinations.

4.1.2.1 First article inspection. Prior to regular production, a sample quantity as specified in Table II shall be subjected to first article tests. The sample(s) shall be manufactured using the same material, equipment, processes, and procedures as will be used in regular production.

4.1.2.1.1 Examination and test. The first article test samples shall be examined for conformance with all of the test requirements of Table II. Failure to meet the requirements specified herein shall be cause for rejection of the first article sample.

4.1.2.1.2 Test failure. Failure of a test sample to comply with the applicable examinations and tests shall result in rejection of the sample. In the event of rejection, corrective action shall be taken and a new preproduction quantity, or portion thereof, shall be submitted in accordance with the procedures specified in the contract or purchase order (see 6.2.1).

4.1.2.2 Propellant moisture and volatiles. During the LAP operation, an in-process inspection shall be conducted to verify that the propellant moisture and volatile content shall conform to the level specified in the specification referenced on the propellant drawing. As a minimum, this inspection shall be conducted on one sample randomly selected at a frequency of once per shift.

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4.2 Quality conformance inspection.

4.2.1 Production lot. The production lot shall be not less than 25,000 cartridges and not more than 100,000 cartridges. A lot shall be defined as:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the referenced drawings and specifications.

b. Cartridge cases, propellants, primers, projectile body assemblies, and boosters having one manufacturer's symbol and one interfix number.

4.2.1.1 Lot inspection data.

4.2.1.1.1 Lot data. A certificate that includes the following information shall be included with each lot.

a. A statement that the lot complies with all the quality assurance provisions specified herein.

b. Number of items inspected.

c. Results obtained, by code number, for all inspections performed.

d. Drawing, specification number, and date, together with an identification and date of change.

e. Number of items in the lot.

f. Data required by 4.1.1.3.7.

g. Date submitted.

4.2.1.1.2 Acceptance data. Acceptance test data shall be reviewed and the DD Form 250 shall be properly signed and stamped by the contractor and procuring activity representative prior to shipment of the ammunition. The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2.1.2 Rejected lots. If a lot is rejected, disposition of the lot shall be as specified in the contract or purchase order (see 6.2.1).

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4.2.2 Lot acceptance tests.

4.2.2.1 Physical examination. The examinations listed in Table V shall be performed on each lot (see 4.2.1). Inspection for critical defects (and major defects, when specified herein) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II or MIL-STD-1235 Inspection Level II, normal inspection unless otherwise specified herein. Individual defects shall have an Acceptable Quality Level (AQL) as specified in the classification of defects. In the event of rejection of a lot, desposition of the lot shall be as specified in the applicable contract or purchase order (see 6.2.1).

4.2.2.2 Functional tests. The tests listed in Table II shall be performed on each lot in accordance with the test methods specified in 4.3. The lot acceptance test results shall be recorded in the uncorrected form as well as after correction for reference data. For acceptance, the corrected test results shall comply with the applicable requirement(s). Acceptance test data shall be included for each individual cartridge tested and also summarized on the format shown on Figure 1, or equivalent.

4.2.2.2.1 Retest provisions.

4.2.2.2.1.1 Velocity, pressure and action time. If a lot does not meet the requirements specified in Table II, a provision to retest that lot is permitted, using the next higher level sample size in accordance with Table V as applicable.

Table V. Classification of defects. 1/

Cartridge, API (Drawing 1397AS190)		
Categories	Defects	Method of Inspection
Critical	100 percent 2/ Primer above flush	 Gage
Major	100 percent 2/ Crack or split in case Crack or split in projectile Chamber interface (Drawing 1395AS160) Maximum length Crimp missing or incomplete 4/ Low propellant charge 5/	 Visual Visual Gage 3/ Gage Visual or gage Gage

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Table V. Classification of defects. 1/ - (Continued)

Cartridge, API (Drawing 1397AS190)		
Major	0.40 AQL Crack or split in primer Loose primer Damaged case Loose windscreen	 Visual Visual, Test Visual Test
Minor	0.65 AQL Primer stake missing	 Visual
Minor	2.50 AQL Marking incorrect, incomplete, illegible or missing Workmanship Case protective coating damage or incomplete	 Visual Visual Visual

Table V Notes:

1/ Visual standards established and approved by the contractor shall be used in evaluating cartridge case and cartridge visual defects. Major components shall have previously been accepted in accordance with the acceptance criteria defined in the specification listed on the applicable drawing.

2/ Inspection verification of prior 100 percent inspections shall be performed in accordance with MIL-STD-105, AQL = 0.25, minimum sample size = 315 with zero defects allowed.

3/ The maximum force to insert the cartridge into the gage shall comply with either of the following criteria:

a. Apply a static force not greater than 265 N.

b. Apply a dynamic force by hand (no mechanical aid) that does not damage the cartridge.

4/ The lot shall not be penalized during inspection verfication for an oversize condition provided that:

a. Two locations 180 degrees apart meet the specified maximum diameter.

b. The crimp is visually complete for 360 degrees.

c. The discrepant cartridge meets the extraction requirements of 3.2.1.5.

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Table V Notes: (Continued)

5/ A cartridge weighing less than the value determined shall be rejected. The inspection method may be by verification of weight of each cartridge or by probe for a fixed minimum propellant height in a charged cartridge case. If the probe is used, the following method shall be used to establish the minimum height:

a. Establish the mean charge weight in accordance with 4.3.1.9.

b. Charge a minimum of 25 cases to the specified weight, measure each case and calculate the mean propellant height.

Inspection verification shall be determined by weighing the total cartridge assembly.

6/ Subject to procuring activity approval, inspection of critical, major and minor characteristics may be performed 100 percent by a qualified and periodically verified probe in lieu of sampling the product utilizing one of the sampling procedures defined above.

4.2.2.2.1.2 Function and casualty. If defects exceed the initial acceptance sample limits of Table I and Table VI, a second sample of 200 cartridges with the accept/reject limits of Table I and Table VI for function and casualty only.

4.2.2.2.1.3 Other retest provisions. No retest provisions are made for accuracy, projectile extraction, projectile torque, propellant weight or waterproofness. Disposition of the lot shall be as specified in the contract or purchase order (see 6.2.1).

4.2.3 Packaging, packing and marking inspection. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable.

4.3 Test methods and procedures. The contractor shall make available to the Government copies of all test methods and procedures to be utilized on a specific contract 30 days prior to initiation of testing. The Government reserves the right to accept all test procedures. The Government shall be notified of any changes to procedures required in the course of testing.

4.3.1 Performance. When performance testing is delegated to a subcontractor or vendor, the Government reserves the right to review and accept all test procedures.

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Table VI. Lot acceptance test function and casualty firing defect. 1/

Accept/Reject Criteria (Double Sampling Plan) 2/							
Lot Size	Cum 3/ Sample Size	Acceptable Quality Levels					
		0.065	0.25	0.40	0.65	1.0	2.5
		Acc/Rej	Acc/Rej	Acc/Rej	Acc/Rej	Acc/Rej	Acc/Rej
22,001/110,000	150 350	0/1 *	0/2 2/3	0/3 4/5	1/4 6/7	2/6 8/9	6/10 16/17

Table VI Notes:

1/ Quantities shown are total of 21, -54, and 71°C tests as specified in Table I and Table II.

Initial test sample size (for lot size of 100,000 units)

+21°C Test Sample Size.....	100
-54°C Test Sample Size.....	25
+71°C Test Sample Size.....	25
Total	150

2/ Includes accept/reject criteria for initial test and cumulative accept/reject criteria for retest.

3/ Retest sample size (in accordance with 4.2.2.2.1.1 and Note 2 of Table V)

+21°C Test Sample Size.....	150
-54°C Test Sample Size.....	25
+71°C Test Sample Size.....	25
Total	200

Initial test and retest sample data are cumulative

150 + 200 = 350

4/ Asterisk (*) indicates single sample plan, (i.e., no retest)

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4.3.1.1 Chamber pressure. The requirement of 3.2.1.1 shall be verified by measuring the pressure at the case mouth. The sample cartridges shall be conditioned at -54, 21, and 71°C and fired from the single-shot test barrel. A piezoelectric-type gage shall be used to measure the chamber pressure. The mean corrected pressure shall be utilized for verification of the requirements of 3.2.1.1.

$$P_{corrected} = P_{case\ mouth} + 20\ MPa + \Delta P$$

4.3.1.2 Action time. The sample cartridges shall be conditioned at -54, 21, and 71°C and fired from the single-shot barrel. The time between firing pin-primer contact and the exit from the muzzle of the projectile shall be measured.

$$t_{corrected} = t_{measured} + \Delta t$$

The corrected action time ($t_{corrected}$) shall comply with the requirements of 3.2.1.2.

4.3.1.3 Velocity. The sample cartridges shall be conditioned at -54, 21, and 71°C and fired from the single-shot test barrel. The velocity shall be measured by two screens or coils located a known distance apart (not less than 20 meters (m)).

A velocity adjustment (V_c) shall be computed utilizing a factor of 0.40 m/s of flight.

$$V_c = 0.40 \cdot X_c$$

where X_c is the distance from muzzle to measurement location

$$V_{corrected} = V_{measured} + V_c + \Delta V$$

The corrected velocity shall comply with the requirements of 3.2.1.3.

4.3.1.4 Accuracy. The test shall be conducted using the single-shot test barrel and a test stand acceptable to the procuring activity. The test cartridges shall be conditioned at 21°C. After firing three barrel-warmer cartridges, the sample cartridges shall be fired without moving either the target or the test barrel. The X and Y coordinates of each impact shall be measured and recorded at a minimum range of 75 m. The standard deviation (σ_x, σ_y) about the mean impact in X and Y shall be calculated. The average standard deviation (σ) shall be calculated as follows:

$$\sigma = (\sigma_x + \sigma_y)/2$$

The requirements of 3.2.1.4 shall be compared with the computed average standard deviation (σ).

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4.3.1.5 Projectile extraction. The sample cartridges shall be tested as specified in AMCR 715-505, Volume 8 except the extraction rate shall be 25.4 ±2.5 mm per minute. The extraction force shall be within the limits specified in 3.2.1.5.

4.3.1.6 Projectile torque. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the cartridge case neck. Torque, to the specified requirement, shall be applied slowly to the projectile, restraining the cartridge case. Movement of the projectile with respect to the cartridge case detectable by misalignment in the scratch mark shall be recorded. The test results shall be examined to determine if the requirements of 3.2.1.6 are met.

4.3.1.7 Penetration. The penetration capability shall be determined in ballistic limit penetration tests against a nominal 38.1 mm armor as specified in MIL-A-12560 at an obliquity of 45.0 ±0.5 degrees. The obliquity is defined as the angle between a line perpendicular to the target surface and the projectile line of flight. The tests shall be conducted on the projectiles that are part of Drawing 1397AS198. The PBL is defined as the average of five penetrations and five nonpenetrations within a velocity of 31 m/s. A penetration is defined as the perforation of the primary armor and a 0.5 mm thick aluminum witness plate placed parallel to and 150 mm behind the armor plate by the penetrator or major portions of the penetrator. The plate shall have a grid of 0.3 m squares and the hardness measured in all squares. The thickness shall be measured for all squares on the plate periphery and interpolated for interior squares. The average hardness and thickness pertaining to the 10 impacts used for the PBL shall be determined. The PBL shall be normalized using the average thickness and hardness from the test to the 38.1 mm plate with a Brinell hardness (BHN) 300 using the penetration equation contained in BAL TR No. 66. Impacts occurring under the following conditions shall be considered unsatisfactory and not utilized in computing the PBL.

a. Projectile yaw greater than 3 degrees.

b. Impacts within 50 mm of the plate edge or a prior impact point.

In order to meet the requirement of 3.2.1.7, the normalized PBL shall be equal to or less than 950 m/s. A minimum of one PBL (5x5) shall be conducted per target plate.

4.3.1.8 Structural integrity. Structural integrity shall be monitored during all single-shot and automatic-gun tests. The integrity of the rotating band (Note 1), the windscreen and the projectile body shall be monitored and evaluated by one or more of the following methods:

a. Radiograph of the projectile during flight

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b. Signature on the yaw card or target

c. Witness panels

d. Soft recovery of the projectile

e. Microflash photographs of the projectile during flight.

Note 1. Rotating band seat fragments having a length less than 10 mm are not considered a structural failure providing band seat knurl imprints are not visible.

4.3.1.9 Waterproofness. The cartridge shall be immersed under at least 50 mm of water (rounds placed horizontally) for a period of 24 hours. The water temperature shall be 21°C. The test items shall meet the requirements of 3.2.1.9.

4.3.1.10 Propellant weight. The charge weight of propellant for a given lot that is required to meet the performance requirements of 3.2.1.1, 3.2.1.2, and 3.2.1.3 shall be determined by ballistic tests in accordance with the procedures specified in the contract or purchase order (see 6.2.1). The charge weight assigned shall be used in determining the minimum cartridge weight for inspection verification of 4.2.2.1.

4.3.1.11 Function and casualty. The test shall be conducted using the AV-8/25 mm Gun System mounted on a ground fire test stand acceptable to the procuring activity.

4.3.1.11.1 First article function and casualty. The AV-8/25 mm Gun System shall be loaded with a complement of 300 cartridges and fired in seven 1-second bursts with a 5-minute cooling period between each burst, the seventh burst shall be 1 record or fire out whichever is less, at a firing rate of 3600 +400, -200 shots per minute (spm). This firing sequence shall be repeated until all cartridges in the test sample are fired. In addition, at least one complement of the test sample shall be a 300 round fire out burst completed at 3600 +400, -200 spm rate. The test data shall be examined to determine that the requirement of 3.2.1.10 is met.

4.3.1.11.2 Lot acceptance function and casualty. The AV-8/25 mm Gun System shall be loaded with the lot test sample cartridges and fired in four 1-second bursts with a 5-minute cooling period between each burst. The fourth burst shall be a fire out, at a firing rate of 3600 +400, -200 spm. The test data shall be examined to determine that the requirement of 3.2.1.10 is met.

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4.3.1.12 Environmental tests. Following exposure to the specified environmental conditions, the test item shall be visually examined for evidence of deterioration and subjected to single-shot tests at 21°C. The cartridge shall satisfy the requirements of 3.2.1.1, 3.2.1.2, 3.2.1.3, 3.2.1.4, and 3.2.1.7 at 21°C.

4.3.1.12.1 Temperature shock. The cartridge shall be subjected to the temperature shock test of MIL-STD-810, Method 503, Procedure I. After exposure, the test cartridges shall be examined and subjected to single-shot tests. The cartridges shall meet the requirements of 3.2.2.1.

4.3.1.12.2 Temperature, humidity and altitude. The cartridge shall be subjected to the temperature-humidity-altitude test of MIL-STD-810, Method 518, Procedure I. After exposure, the test cartridges shall be examined and subjected to a single-shot tests. The cartridges shall meet the requirements of 3.2.2.2.

4.3.1.12.3 Salt fog. The cartridge shall be subjected to the salt fog test of MIL-STD-810, Method 509, Procedure I. After exposure, the test cartridges shall be examined and subjected to single-shot tests. The cartridges shall meet the requirements of 3.2.2.3.

4.3.1.13 Propellant contamination. The propellant from each of the cartridges tested for projectile extraction will be examined visually for contamination. The propellant shall meet the requirements of 3.3.2.3.

4.3.1.14 Reliability. The reliability of the cartridge shall be computed using all applicable data gathered during the tests of Table I and Table II. The requirement of 3.2.3 in satisfied if no failures as outlined in 3.2.3 occur. Nonchargeable failures defined in 4.3.1.14.1 shall not be included in the reliability calculations.

4.3.1.14.1 Nonchargeable failure. Failure attributed to any of the following shall not be considered a failure of the cartridge for the purposes of reliability measurement:

a. Failure of test equipment.

b. Failure caused by personnel errors.

c. Failure of associated equipment.

d. Dependent (secondary) failure.

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5. PACKAGING

5.1 Preservation, packaging and packing. The cartridge shall be preserved and packed in accordance with the requirements of the Transportation Packaging Order (TPO) Drawing established by the Government (see 6.2.1). Packaging and packing shall conform to all applicable requirements for the shipment of hazardous items as specified in 49 CFR 100-199.

5.2 Marking. Marking for shipment shall be as specified in the contract and shall comply with MIL-STD-129. Exterior containers shall show the caution markings specified in 49 CFR 100-199, indicating the appropriate classification of ammunition.

5.3 Documentation with shipment. Each shipment of ammunition shall contain data cards in accordance with MIL-STD-1167, lot number assignment in accordance with such items as case, primer, ignition system, propellant, charge weights, projectile, and lot identification.

5.4 Shipping. When shipments of more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent moving in transit.

6. NOTES

6.1 Intended use. The PGU-20/U Armor Piercing Incendiary Cartridge is intended for firing in the GAU-12/U gun and is to be compatible with the M242 and the KBA-B02 guns.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Lot acceptance test requirement including cartridges for Government evaluation, if required (see 4.2.2).

c. Disposition of rejected acceptance lots (see 4.2.1.2 and 4.2.2.1).

d. Packing and shipping instructions (see 5.1 and 5.4).

e. Misfire not due to primer (see Table I).

f. Unlisted firing defects (see 4.1.1.3.7).

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g. Test sample failure (see 4.1.2.1.2).

h. Lot retest provisions (see 4.2.2.2.1.3)

i. Lot propellant weight (see 4.3.1.10).

6.2.2 Data requirement. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.	Option
5.3	Data cards	DI-E-2001	---
Table I	Failure Report	UDI-R-21141A	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist units as specified in 4.1.2.1. The procuring activity should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the document's first article.

6.3 Reference conversion chart. The conversion chart from Metric Units to U.S. Customary Units provided in Table VII is for reference only.

Preparing activity:
NAVY(AS)

Project (1305-N001)

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Table VII. Reference conversion chart (approximate).

°C	°F	mm	in	m	ft
-54	-65	0.5	0.02	0.3	0.98
-40	-40	2.0	0.08	20	65.6
3	37	2.5	0.1	75	246.1
5	41	5	0.2	80	262.5
18	64	10	0.394	100	328.1
21	70	25.4	1.0	200	656.2
24	75	36.5	1.4	750	2461
71	160	38.1	1.5	1500	4921
300	572	50.0	1.969	3048	10000
		67.4	2.654	5944	19501
		90.8	3.6		
		150	5.906		

MPa	kpsi	m/s	ft/s	kg/m^3	lb/ft^3
20	2.9	0.4	1.3	1.5	0.094
25	3.6	3	9.8	1.6	0.100
30	4.4	10	32.8		
425	61.6	15	49.2		
476	69.4	30	98.4		
		31	101.7		
		950	3116.8		
		1000	3280.8		

N·m	ft/lbs	$(m/s^2)^2/Hz$	$(g)^2/Hz$	m/s^2	ft/s^2
3.4	2.51	3.8	0.04	196	643

grams	lbs	N	lbs	degrees	deg min sec
3	0.007	265	60	0.5	0° 30'
94	0.21	8000	1800	5.5	5° 30'
214	0.47	16900	3800		
224	0.49			7.5	7° 30'
432	0.95				
532	1.17				
540	1.19				

$kg.m^2$	$lb\ in^2$
2.3×10^{-3}	7.86
7.7×10^{-5}	0.26
8.5×10^{-5}	0.29
9.2×10^{-6}	0.03

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LOT NUMBER	PART NUMBER	REVISION
QUANTITY	CONTRACT	SPECIFICATION
DATE TESTED	QUANTITY TESTED	

COMPONENTS

PART NAME	PART NUMBER	MANUFACTURER	LOT NUMBER
Cartridge Case			
Cartridge Cases Primed			
Primer			
Propellant			
Projectile Assembly			
Barrel			
Reference Rounds			

TEST		21°C	-54°C	+71°C			WTRPRF
VELOCITY (m/s)	MEAN (X)						
	STANDARD DEVIATION (σ)						
PRESSURE (MPa)	MEAN (X)						
	STANDARD DEVIATION (σ)						
	+ 3 σ						
ACTION TIME (ms)	MEAN						
	Maximum						

ACCURACY	σx (Mils)		DEFECTS
	σy (Mils)		
	R (Milliradians)		
PROJECTILE EXTRACTION (N)	Minimum		
	Maximum		
	Mean		
PROPELLANT CHARGE (Info) (Grams)	(Grams)		
	Range		WAIVERS OR DEVIATIONS
PROPELLANT CONTAMINATION			
FUNCTION AND CASUALTY			
TORQUE			
CARTRIDGE WEIGHT (Grams)	Mean ( )		
	Standard Deviation (σ)		

APPROVALS

NAME	ORGANIZATION	DATE
NAME	ORGANIZATION	DATE

Figure 1. Ballistic test data summary.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85624(AS)	CARTRIDGE, 25 MILLIMETER, PGU-20/U (API)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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